

MATH3881/5881: Statistical Machine Learning Theory

Week 4: Optimisation in Machine Learning: Algorithms — Exercises and Solutions

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Exercises and Solutions

Exercise 4.0.1 (Variance under without-replacement sampling)

Fix $\theta \in \mathbb{R}^d$, write $g_i := \nabla C_i(\theta)$ and $\bar{g} := \frac{1}{n} \sum_i g_i = \nabla C(\theta)$, and let $\Sigma := \text{Var}(g_{I_{t,1}})$ when $I_{t,1}$ is uniform on $\{1, \dots, n\}$, equivalently $\Sigma = \frac{1}{n} \sum_i (g_i - \bar{g})(g_i - \bar{g})^\top$. Let $\mathcal{I}_t = (I_{t,1}, \dots, I_{t,n_b})$ be a uniform size- n_b subset of $\{1, \dots, n\}$ drawn *without replacement* within a single mini-batch, and $\hat{g}^{(t)} := \frac{1}{n_b} \sum_{i \in \mathcal{I}_t} g_i$. Show

$$\text{Var}(\hat{g}^{(t)}) = \frac{1}{n_b} \cdot \frac{n - n_b}{n - 1} \Sigma.$$

Solution. *Step 1.* (Expansion.) Writing $\hat{g}^{(t)} - \bar{g} = \frac{1}{n_b} \sum_{j=1}^{n_b} (g_{I_{t,j}} - \bar{g})$ and using $\mathbb{E}g_{I_{t,j}} = \bar{g}$,

$$\text{Var}(\hat{g}^{(t)}) = \frac{1}{n_b^2} \sum_{j,k=1}^{n_b} \text{Cov}(g_{I_{t,j}}, g_{I_{t,k}}),$$

with $\text{Cov}(\xi, \eta) := \mathbb{E}[(\xi - \mathbb{E}\xi)(\eta - \mathbb{E}\eta)^\top]$.

Step 2. (Diagonal terms, $j = k$.) Each $I_{t,j}$ has marginal uniform distribution on $\{1, \dots, n\}$ (the marginal of a uniform subset's j -th element), so $\text{Cov}(g_{I_{t,j}}, g_{I_{t,j}}) = \text{Var}(g_{I_{t,j}}) = \Sigma$. There are n_b such terms, contributing $n_b \Sigma$.

Step 3. (Off-diagonal terms, $j \neq k$.) Under without-replacement sampling, $\mathbb{P}(I_{t,j} = a, I_{t,k} = b) = 1/[n(n-1)]$ for $a \neq b$, so

$$\mathbb{E}[g_{I_{t,j}} g_{I_{t,k}}^\top] = \frac{1}{n(n-1)} \sum_{a \neq b} g_a g_b^\top.$$

Using $\sum_{a \neq b} g_a g_b^\top = (\sum_a g_a)(\sum_b g_b)^\top - \sum_a g_a g_a^\top = n^2 \bar{g} \bar{g}^\top - n(\Sigma + \bar{g} \bar{g}^\top)$, where the last step uses $\Sigma = \frac{1}{n} \sum_a g_a g_a^\top - \bar{g} \bar{g}^\top$,

$$\mathbb{E}[g_{I_{t,j}} g_{I_{t,k}}^\top] = \frac{(n^2 - n) \bar{g} \bar{g}^\top - n \Sigma}{n(n-1)} = \bar{g} \bar{g}^\top - \frac{\Sigma}{n-1}.$$

Hence $\text{Cov}(g_{I_{t,j}}, g_{I_{t,k}}) = \mathbb{E}[g_{I_{t,j}} g_{I_{t,k}}^\top] - \bar{g} \bar{g}^\top = -\Sigma/(n-1)$. There are $n_b(n_b - 1)$ such ordered pairs, contributing $-n_b(n_b - 1) \Sigma/(n-1)$.

Step 4. (Combine.) Adding the two contributions and dividing by n_b^2 ,

$$\text{Var}(\hat{g}^{(t)}) = \frac{1}{n_b^2} \left[n_b \Sigma - \frac{n_b(n_b - 1)}{n - 1} \Sigma \right] = \frac{\Sigma}{n_b} \left[1 - \frac{n_b - 1}{n - 1} \right] = \frac{\Sigma}{n_b} \cdot \frac{n - n_b}{n - 1}. \quad \square$$

Exercise 4.0.2 (Momentum with constant gradient)

Suppose $\nabla C(\theta^{(\tau)}) = g$ is constant for all $\tau \geq 0$, with $v^{(0)} = 0$. Using the unrolled form $v^{(t+1)} = (1 - \beta) \sum_{\tau=0}^t \beta^{t-\tau} \nabla C(\theta^{(\tau)})$, find $v^{(t+1)}$ in closed form. What does $v^{(t+1)}$ converge to as $t \rightarrow \infty$, and what does the momentum update $\theta^{(t+1)} = \theta^{(t)} - \alpha v^{(t+1)}$ reduce to in this limit?

Solution. *Step 1.* (Substitute the constant gradient.)

$$v^{(t+1)} = (1 - \beta)g \sum_{\tau=0}^t \beta^{t-\tau}.$$

Step 2. (Evaluate the geometric sum.) Reindex with $k = t - \tau$:

$$\sum_{\tau=0}^t \beta^{t-\tau} = \sum_{k=0}^t \beta^k = \frac{1 - \beta^{t+1}}{1 - \beta}.$$

Hence $v^{(t+1)} = (1 - \beta^{t+1})g$.

Step 3. (Limit.) Since $\beta \in [0, 1)$, $\beta^{t+1} \rightarrow 0$ as $t \rightarrow \infty$, so $v^{(t+1)} \rightarrow g$.

Step 4. (Limit of the update.) The momentum step becomes, in the limit,

$$\theta^{(t+1)} \rightarrow \theta^{(t)} - \alpha g,$$

i.e. a standard gradient descent step with step size α on the constant gradient. Momentum reduces to plain GD when the gradient is constant; its benefit appears in *transient* phases where the gradient changes direction, where it damps oscillations across narrow valleys and builds up speed along consistent descent directions.

Exercise 4.0.3 (Adaptive Methods: Bias Correction and Effective Rates)

- (a) With $t = 0$, $v^{(0)} = s^{(0)} = 0$, gradient $g = \nabla C(\theta^{(0)})$: compute $v^{(1)}, s^{(1)}, \hat{v}^{(1)}, \hat{s}^{(1)}$ symbolically, and show that for any non-zero coordinate g_i , $|\theta_i^{(1)} - \theta_i^{(0)}| = \alpha/(1 + \varepsilon/|g_i|) \approx \alpha$.
- (b) For a 1-d problem with constant gradient $g > 0$ from $s^{(0)} = 0$, derive closed-form $s^{(t)}$ for AdaGrad and RMSProp, and show the effective step sizes decay as $\alpha/(g\sqrt{t}) \rightarrow 0$ and converge to α/g respectively.

Solution.

- (a) *Step 1.* (One Adam iteration from zero initial moments.) With $v^{(0)} = 0$, $s^{(0)} = 0$ and gradient g ,

$$v^{(1)} = \beta \cdot 0 + (1 - \beta) g = (1 - \beta) g, \quad s^{(1)} = \gamma \cdot 0 + (1 - \gamma) (g \odot g) = (1 - \gamma) g^2 \text{ (element-wise).}$$

Step 2. (Bias correction at $t = 0$: divide by $1 - \beta^1 = 1 - \beta$ and $1 - \gamma^1 = 1 - \gamma$.)

$$\hat{v}^{(1)} = \frac{(1 - \beta) g}{1 - \beta} = g, \quad \hat{s}^{(1)} = \frac{(1 - \gamma) g^2}{1 - \gamma} = g^2.$$

Step 3. (Parameter update for coordinate i .) The Adam update gives

$$\theta_i^{(1)} - \theta_i^{(0)} = -\alpha \frac{\hat{v}_i^{(1)}}{\sqrt{\hat{s}_i^{(1)} + \varepsilon}} = -\alpha \frac{g_i}{|g_i| + \varepsilon}.$$

Hence

$$|\theta_i^{(1)} - \theta_i^{(0)}| = \frac{\alpha |g_i|}{|g_i| + \varepsilon} = \frac{\alpha}{1 + \varepsilon/|g_i|} \approx \alpha$$

whenever $\varepsilon \ll |g_i|$. Bias correction ensures every coordinate receives a step of roughly $\pm\alpha$ at the very first iteration, regardless of the gradient's scale. Without bias correction, the same calculation gives first-step magnitude $\alpha(1 - \beta)/\sqrt{1 - \gamma}$, which for Adam defaults $\beta = 0.9$, $\gamma = 0.999$ equals $\alpha \cdot 0.1/\sqrt{0.001} \approx 3.16\alpha$, so the uncorrected initial step would be about $3\times$ too large and could destabilise training.

- (b) **AdaGrad** ($s^{(0)} = 0$, constant gradient g). The accumulator is

$$s^{(t)} = \sum_{\tau=0}^{t-1} g^2 = t g^2.$$

Effective step size: $\alpha/\sqrt{s^{(t)}} = \alpha/(g\sqrt{t}) \rightarrow 0$ as $t \rightarrow \infty$.

RMSProp ($s^{(0)} = 0$, decay $\gamma \in [0, 1)$). Unrolling the recursion,

$$s^{(t)} = (1 - \gamma) \sum_{k=0}^{t-1} \gamma^k g^2 = g^2 (1 - \gamma^t).$$

Hence $s^{(t)} \rightarrow g^2$ as $t \rightarrow \infty$, and the effective step size $\alpha/\sqrt{s^{(t)}} \rightarrow \alpha/g$, a non-zero constant. RMSProp maintains a useful learning rate indefinitely; AdaGrad's rate decays to zero and the method eventually stalls.

Exercise 4.0.4 (Newton's Method: Convergence on an Explicit Function)

Let $C(\theta) = e^\theta - 2\theta$.

- (a) Find θ^* exactly. From $\theta^{(0)} = 0$, perform two Newton steps and report errors $e_t = |\theta^{(t)} - \theta^*|$ for $t = 0, 1, 2$.
- (b) Compute $\kappa_C = |C'''(\theta^*)|/(2|C''(\theta^*)|)$, check the bound $e_1 \lesssim \kappa_C e_0^2$, and compare ratios e_1/e_0^2 and e_2/e_1^2 .
- (c) From $\theta^{(0)} = 0$ and $\theta^{(1)} = 1$, perform one secant step and compare the error with Newton's.

Solution.

- (a) $C'(\theta) = e^\theta - 2$, $C''(\theta) = e^\theta > 0$, $C'''(\theta) = e^\theta$.

Minimiser: $C'(\theta^*) = 0 \Rightarrow e^{\theta^*} = 2 \Rightarrow \theta^* = \ln 2 \approx 0.693$.

Newton step 1 from $\theta^{(0)} = 0$:

$$\theta^{(1)} = 0 - \frac{e^0 - 2}{e^0} = 0 - \frac{-1}{1} = 1.$$

Newton step 2 from $\theta^{(1)} = 1$:

$$\theta^{(2)} = 1 - \frac{e^1 - 2}{e^1} = \frac{2}{e} \approx 0.736.$$

Errors:

$$e_0 = \ln 2 \approx 0.693, \quad e_1 = |1 - \ln 2| = 1 - \ln 2 \approx 0.307, \quad e_2 = \left| \frac{2}{e} - \ln 2 \right| \approx 0.043.$$

- (b) At $\theta^* = \ln 2$: $C''(\theta^*) = e^{\ln 2} = 2$ and $C'''(\theta^*) = 2$. So

$$\kappa_C = \frac{|C'''(\theta^*)|}{2|C''(\theta^*)|} = \frac{2}{2 \times 2} = \frac{1}{2}.$$

Theoretical bound: $e_1 \lesssim \kappa_C e_0^2 = \frac{1}{2}(\ln 2)^2 \approx \frac{1}{2}(0.693)^2 \approx 0.240$. Observed $e_1 \approx 0.307 > 0.240$, so the bound *does not hold* at $t = 1$. This is expected: the quadratic convergence bound is only an asymptotic result valid when e_t is small (close to θ^*). At $e_0 \approx 0.693$, we are still in the non-asymptotic regime where higher-order terms in the Taylor expansion matter.

Checking the convergence ratios:

$$\frac{e_1}{e_0^2} \approx \frac{0.307}{0.480} \approx 0.64, \quad \frac{e_2}{e_1^2} \approx \frac{0.043}{0.094} \approx 0.46.$$

The ratio $e_2/e_1^2 \approx 0.46$ is closer to $\kappa_C = 0.5$ than $e_1/e_0^2 \approx 0.64$ is, because $e_1 \approx 0.307$ is smaller, bringing us closer to the asymptotic regime. Both ratios approach κ_C as the iterate converges to θ^* .

(c) Secant step from $\theta^{(0)} = 0$, $\theta^{(1)} = 1$:

$$C'(\theta^{(0)}) = e^0 - 2 = -1, \quad C'(\theta^{(1)}) = e - 2 \approx 0.718.$$

Secant approximation to C'' :

$$C''(\theta^{(1)}) \approx \frac{C'(\theta^{(1)}) - C'(\theta^{(0)})}{\theta^{(1)} - \theta^{(0)}} = \frac{(e - 2) - (-1)}{1} = e - 1 \approx 1.718.$$

Secant step:

$$\theta_{\text{sec}}^{(2)} = 1 - \frac{e - 2}{e - 1} = \frac{(e - 1) - (e - 2)}{e - 1} = \frac{1}{e - 1} \approx \frac{1}{1.718} \approx 0.582.$$

Error: $e_2^{\text{sec}} = |1/(e - 1) - \ln 2| \approx |0.582 - 0.693| \approx 0.111$.

Comparing: $e_2^{\text{Newton}} \approx 0.043$ versus $e_2^{\text{sec}} \approx 0.111$. Newton is about $2.6\times$ more accurate at this step. This is consistent with Newton's quadratic convergence (rate ≈ 2) versus the secant method's superlinear convergence (rate ≈ 1.618). The advantage of the secant method is that it requires only first-derivative evaluations, not second derivatives.

Exercise 4.0.5 (Quasi-Newton: Secant Equation and Memory)

Let $C(\theta) = \frac{1}{2}\theta^\top A\theta - b^\top \theta$ with $A \succ 0$.

- (a) Show $g_s^{(t)} = A\Delta^{(t)}$ and state what the secant equation requires of B_{t+1} .
- (b) Verify the curvature condition $g_s^{(t)\top} \Delta^{(t)} > 0$ holds for $\Delta^{(t)} \neq 0$.
- (c) For $d = 10^7$ parameters and 32-bit floats, compare the memory for BFGS ($B_t \in \mathbb{R}^{d \times d}$) and L-BFGS ($m = 10$ curvature pairs) in GB.

Solution.

(a) The gradient of C is $\nabla C(\theta) = A\theta - b$. The gradient difference is:

$$g_s^{(t)} = \nabla C(\theta^{(t+1)}) - \nabla C(\theta^{(t)}) = A\theta^{(t+1)} - A\theta^{(t)} = A\Delta^{(t)}.$$

Substituting into the secant equation $B_{t+1}g_s^{(t)} = \Delta^{(t)}$:

$$B_{t+1}A\Delta^{(t)} = \Delta^{(t)}.$$

For this to hold for *all* $\Delta^{(t)} \neq 0$, we need $B_{t+1}A = I$, i.e. $B_{t+1} = A^{-1}$ (the exact inverse Hessian). In other words, the quasi-Newton update is consistent: a perfect approximation $B_{t+1} = H^{-1}$ satisfies the secant equation exactly. The BFGS update incrementally corrects B_t toward A^{-1} using each new step/gradient pair.

(b) The curvature condition:

$$g_s^{(t)\top} \Delta^{(t)} = (A\Delta^{(t)})^\top \Delta^{(t)} = (\Delta^{(t)})^\top A \Delta^{(t)} > 0$$

for all $\Delta^{(t)} \neq 0$, since A is positive definite by assumption. Positive definiteness of A (i.e. strong convexity of C) guarantees the curvature condition, ensuring a valid positive definite update at every step.

(c) Memory calculation with $d = 10^7$ parameters (4 bytes per float):

- **BFGS**: Stores $B_t \in \mathbb{R}^{d \times d}$.

$$d^2 \times 4 \text{ bytes} = (10^7)^2 \times 4 = 4 \times 10^{14} \text{ bytes} = 4 \times 10^5 \text{ GB} = 400,000 \text{ GB}.$$

- **L-BFGS**: Stores $2m$ vectors in $\mathbb{R}^d$ (the pairs $\Delta^{(k)}$ and $g_s^{(k)}$):

$$2 \times 10 \times 10^7 \times 4 = 8 \times 10^8 \text{ bytes} = 0.8 \text{ GB}.$$

BFGS would require 400,000 GB — roughly 500 times the total memory of a high-end server cluster. L-BFGS requires a mere 0.8 GB, well within the capacity of a single GPU. This is why BFGS is completely infeasible for modern deep learning: $d \sim 10^7$ – 10^9 parameters make $O(d^2)$ storage impossible. L-BFGS ($O(md)$ storage with small m) remains viable for moderate-scale problems or as a fine-tuning step after stochastic pre-training.